

**A 40yo requires a laparotomy 10 days after an isolated traumatic spinal cord transaction at C6. (a) outline the key anaesthetic issues (50%)
(b) how would these influence your anaesthetic management (50%)**
(Modified from MMed 2016, ANZCA 2015A, 2013B)

Examiner Report

Candidates were expected to mention

- A. Key issues including unstable c spine, neurogenic shock and acute laparotomy
- B. Anaesthetic management would be expected to include a sensible airway plan, managing the neurogenic shock and dealing with an unfasted, unwell patient for laparotomy

Sample Answer

This is a middle-aged patient with an isolated cervical spinal cord transaction in the early hyperreflexia phase (of spinal shock) requiring a laparotomy, which may be urgent in nature.

Key anaesthetic issues	Influence on my anaesthetic management
Relating to C-Spine Injury	
Severity of spinal cord injury (SCI) • Patient should be assessed for presence of <u>neurogenic shock</u> ○ This is more likely if SCI is <u>complete (vs incomplete)</u> based on ASIA scoring. Distributive shock may last for weeks, and should be supported accordingly. • Presence of <u>sympathetic denervation</u> may predispose patient to haemodynamic instability. • Intra-op hypovolaemia from <u>blood loss</u> is <u>poorly tolerated</u> ○ Pre-existing <u>vasodilatory shock</u> and <u>bradycardia</u> (loss of cardio-accelerator fibres), compounded by <u>hypovolaemia</u> → <u>Refractory hypotension</u> and impaired end organ perfusion • Given that it is Day 10, patient is likely in the <u>early hyperreflexia phase (of spinal shock)</u>. ○ <u>Autonomic dysreflexia</u> is unlikely to be present at D10 of injury (usually after months-year).	• Ensure 2 <u>large bore IV cannulas</u>, with <u>fluid preload</u> of 500-1000ml of IV crystalloids before induction. • Resuscitation drugs, including <u>vasopressors and inotropes</u> should be prepared and ready for use in OT. • <u>IA line with Flotrac</u> to guide haemodynamics and fluid management. <u>KIV CVP line</u> for measurement of SVRI, or if vasopressors/ inotropes are required. • <u>IV Atropine 300mcg</u> pre-induction to reduce risks of severe bradycardia/ asystole during intubation. • <u>Neurological status</u> should be checked and documented <u>pre- and post-op</u>, especially if SCI is incomplete, with residual sensory/motor function • <u>Post-op HD bed</u> should be available prior to operation.
Risk of secondary SCI injury (including extension of injury above C6) if adequate spinal perfusion pressure was not maintained.	• Avoid hypotension, hypoxia, anaemia, hypercarbia, acidosis. • <u>Anaemia</u> should be corrected pre-op. Ensure <u>valid GXM</u>. • <u>Blood products</u> and <u>fluid warmer</u> should be available in OT if surgical bleeding is anticipated. • A <u>higher MAP</u> (> 85mmHg) target may help with maintenance of spinal perfusion pressure
Unstable C-Spine with potentially difficult airway.	• Need for <u>cervical spine immobilisation</u> if no surgical fixation. ○ <u>C-collar</u> should be kept on until airway manipulation

I would check if <u>surgical fixation</u> has been done, and whether there is any <u>recent anaesthesia history, including airway documentation</u>.	<u>Log-roll</u> transfer (requires at least 6 OT staff) from trolley to OT table - Manual in-line stabilisation (<u>MILS</u>) during intubation <u>Head blocks/ sandbags</u> to be placed at patient's head to maintain spinal immobilisation intra-op. - Use of <u>airway adjuncts</u>, including counselling of patient for <u>awake fibreoptic intubation</u> in previously documented difficult airway.
Upregulation of extrajunctional ACh receptors after 48-72h of injury.	<u>Suxamethonium should be avoided</u> in view of risk of hyperkalaemia with administration.
Increased risk of infection <u>Impaired cough</u> from intercostal muscle and accessory muscles weakness predisposes patient <u>pneumonia/atelectasis</u>. This can lead to <u>impaired oxygenation, ventilation, and post-op respiratory complications</u>. - Urinary retention and use of IDC predispose patient to <u>urinary tract infection</u> (UTI).	<u>Pre-op CXR</u> to exclude pneumonia and atelectasis. - Infection should be <u>investigated and treated</u>. - Lung protective ventilatory strategies, use of <u>PEEP</u>, lowest FiO2 to maintain SpO2 > 95%, <u>recruitment</u> manoeuvres, and <u>ETT suctioning</u> intra-op. - Full reversal of neuromuscular blockade, guided by <u>TOF ratio > 0.9</u> - Appropriate counselling of <u>peri-op respiratory complications</u>, including <u>delayed extubation</u> in ICU in event of <u>difficulty weaning ventilation</u> at the end of op.
Impaired thermoregulation Leads to higher risk of hypothermia	<u>Temperature monitoring</u> with temp probe intra-op - Use of <u>passive and active warming</u> devices, including warming blankets and fluid warmers
Prolonged immobilisation - Predisposes patient to pressure sores and peripheral nerve injury (due to insensate limbs) - Predisposes patient to deep vein thrombosis (DVT). Patient may be on anti-coagulant (e.g IV Heparin) Peripheral oedema from prolonged recumbent position can make IV cannulation difficult	- Careful positioning of arms and legs to maintain <u>neutral</u> positions. - Pressure points such as <u>elbows and heels</u> should be <u>padded</u>. - Padded dressings (Mepilex) to <u>protect bony prominences</u> such as coccyx. <u>Heparin</u> (if started for DVT prophylaxis) will need to be <u>stopped 6H</u> pre-op. <u>Calf compressors</u> should be continued intra-op. <u>Ultrasound guidance</u> may be required if IV access is difficult

Increased risk of gastric stress ulcers	• <u>Continue PPI</u> peri-operatively
Relating to surgical pathology requiring laparotomy	
Indication and urgency of surgery should be discussed with surgeon • A patient who requires emergency laparotomy may be <u>unfasted</u> and <u>unwell</u>. • Limited time for optimisation of patient prior to surgery	• <u>Elective procedures should be postponed</u> until patient has recovered from spinal shock. • Patient should be <u>counselled on elevated peri-op MACE risks</u> if emergency op were to proceed.
Aspiration risk in unfasted patient/acute abdomen • This may also be contributed by gastroparesis secondary to SCI.	• Modified RSI ○ Use of <u>Rocuronium 1.2mg/kg with Sugammadex</u> standby ○ <u>Avoid cricoid pressure</u> as SCI is at C6. Alternatively, bimanual cricoid pressure may be applied.
Dehydration/ electrolyte abnormalities • If patient has prolonged NBM or has GI losses.	• <u>Fluid deficit</u> and <u>electrolyte abnormalities</u> should be corrected pre-op.
Surgical pain from laparotomy incision • Complete SCI → Skin will be insensate below level of lesion. Risk of opioid overdose (including respiratory depression) if opioid dosage is not adjusted • Incomplete SCI → Increased risk of transition to chronic pain if surgical pain is not well-managed. Concurrent adjustment disorder/ depression may affect pain perception. • PCA Opioid will not be useful if patient has no upper limb motor function.	• Analgesia will need to be <u>titrated to patient's response</u>. • <u>Use of adjuvants</u> (gabapentin/ ketamine) may be useful for patient with incomplete SCI. • Use of <u>regional anaesthesia</u> (e.g. bilateral TAP blocks) or <u>local infiltrative technique</u> (On-Q painbuster) to reduce opioid use. • <u>Nurse controlled Analgesia (NCA)</u> may be needed in place of PCA opioids.